

ARAV MEHTA

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Education

University of Massachusetts Amherst | GPA: 3.12

Expected Graduation - May 2027

B.S. in Computer Science

Amherst, MA

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Web Programming, Linear Algebra, Practice and Applications of Data Management.

Technical Skills

Languages: Python, Java, C, JavaScript/TypeScript, SQL.

Cloud/DevOps: AWS (S3, EC2, SageMaker, IAM, Lambda, API Gateway, DynamoDB, Cognito, SQS, CloudFront, CloudWatch, X-Ray, CDK), Supabase, Cloudflare Workers.

Tools & Frameworks: React, Next.js, Node.js, Flutter, FastAPI, Prisma, PyTorch, HuggingFace, Scikit-Learn.

Experience

Tortoise Finance

Jun 2025 – Aug 2025

Software Engineering Intern

- Reduced analyst triage time by **55%** (serving $\approx 4k$ users, page-load p95 ≈ 280 ms) by developing an internal portal that replaced spreadsheet/email workflows and centralized intake, approvals, and tracking.
- Lowered API error rate by **88%** and increased peak throughput by **80%** (to $\approx 1.8x$ baseline) by optimizing database access patterns and introducing server-side caching.
- Shortened release rollout time from **12** to **6** minutes while keeping change-failure rate under **5%** by strengthening the CI/CD pipeline with health checks and guarded deploys as the **sole engineer**.
- Reduced Mean Time To Recovery from **42** to **16** minutes by implementing tracing, dashboards, and alerts, enabling faster detection and resolution while maintaining zero-downtime releases.

Outamation

May 2025 – Jun 2025

AI and Automation Extern

- Built a scalable document understanding pipeline using Tesseract OCR and PyMuPDF, reducing manual processing by **80%** across diverse document types.
- Developed and deployed a production-ready RAG pipeline using LlamaIndex and ChromaDB to enable semantic search over large unstructured text sets.
- Enhanced retrieval precision by **40-60%** through query rewriting, semantic chunking, and fusion-based hybrid search, improving search relevance.
- Contributed to technical documentation and iterative design reviews with stakeholders, ensuring project scalability and maintainability.

Projects

Dynamic-RAG (HackRU 2025 Winner – MLH Best Use of Snowflake API) | FastAPI, FastMCP, Next.js, Snowflake Cortex, Gemini, MongoDB, Docker, AWS ([Project Details](#))

- Developed a context MCP system using FastMCP that enabled LLMs to share and sync context in real time across users and platforms, reducing redundant token usage by **23%**.
- Created a Next.js dashboard displaying live context updates, and rollback functionality, visualizing project graphs and team activity.
- Added chunking logic with tagged adjacency-list structures to allow visualization of relationships between context chunks.
- Integrated Gemini APIs for embedding and RAG pipeline generation, improving semantic search accuracy by over **40%**, and used the Snowflake Cortex API for efficient vector storage, similarity search, and real-time observability.

DormDash–Student Food Delivery Platform (Ongoing) | Flutter, Supabase, Cloudflare, PostgreSQL, Stripe

- Independently developing a cross-platform app (Flutter) with a scalable Supabase backend (PostgreSQL, RLS, real-time channels) for secure student-to-student food delivery.
- Implemented .edu-restricted authentication (Google OAuth, email login), prototyping Stripe payments and QR-based courier access with CI/CD-ready deployment.
- Designed a cloud-native architecture leveraging Cloudflare Workers for cost-effective scaling, high availability, and fault tolerance.

Multimodal Sentiment Analyzer | Next.js, React, Tailwind, Auth.js, PyTorch, AWS, Tensorflow ([GitHub](#))

- Built a full-stack SaaS for real-time video sentiment analysis; increased sentiment analysis accuracy by **8%** through optimized hyperparameter tuning and model refinement.
- Engineered automated preprocessing pipelines (FFmpeg, Whisper, OpenCV), integrated S3 uploads, NextAuth.js authentication, and quota-tracked API access via Prisma.

Activities

CICS Peer Mentor | UMass Amherst

Sep 2025 – Present

- Coached CS freshmen through one-on-one sessions by identifying individual challenges and providing targeted resources, resulting in increased sense of belonging and strong positive feedback.